ECE 102 Homework 7

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1. (10 points) Sampling

Assume $x(t)$ a real bandlimited signal where $X(j\omega)$ is non-zero for $|\omega| \leq 2\pi B$ rad/s. If F_s Hz is the Nyquist rate of $x(t)$, determine the Nyquist rate of the following signals in terms of B:

(a) $x(t+1)$

Solution:

$$\mathcal{F}[x(t+1)] = e^{j\omega}X(j\omega) \tag{1}$$

Since the magnitude spectrum stays the same:

$$|e^{j\omega}X(j\omega)| = |e^{j\omega}||X(j\omega)| = |X(j\omega)| \tag{2}$$

the bandwidth stays the same. The bandwidth of $x(t)$ is $\frac{2\pi B}{2\pi} = B$. Thus the Nyquist rate is $2B$ Hz.

(b) $\cos(2\pi Bt)x(t)$

Solution:

$$\mathcal{F}[\cos(2\pi Bt)x(t)] = \frac{1}{2\pi}X(j\omega) * \pi(\delta(\omega - 2\pi B) + \delta(\omega + 2\pi B)) \tag{3}$$

$$= \frac{1}{2}X(j\omega - 2\pi B) + \frac{1}{2}X(j\omega + 2\pi B) \tag{4}$$

Since $X(j\omega)$ is non-zero for $|\omega| \leq 2\pi B$, and we are shifting each by $\pm 2\pi B$, the signal is supported for $|\omega| \leq 4\pi B$ (bandwidth). The Nyquist rate then is $2 \cdot \frac{4\pi B}{2\pi} = 4B$ Hz

(c) $x(t) + x(2t)$

Solution:

$$\mathcal{F}[x(t) + x(2t)] = X(j\omega) + \frac{1}{2}X\left(j\frac{\omega}{2}\right) \tag{5}$$

This ω is supported on $[-4\pi B, 4\pi B$, which means the Nyquist rate is $2 * \frac{4\pi B}{2\pi} = 4B$ Hz.

2. (20 points) Bandpass sampling

The figure below shows the Fourier transform of a real bandpass signal, i.e., a signal whose frequencies are not centered around the origin. We want to sample this signal. Let F_s in Hz represent the sampling frequency.

(a) (5 points) One option is to sample this signal at the Nyquist rate. Then to recover the signal, we pass its sampled version through a low pass filter. What is the Nyquist rate of this signal? What is the cutoff frequency of the low pass filter?

Solution:

The Nyquist rate is double the maximum frequency of $X(j\omega)$, which is $2 \cdot \frac{5\pi}{2\pi} = 5$ Hz. The cutoff frequency of this low pass filer would then be for $|\omega| \leq 5\pi$ or 2.5 Hz.

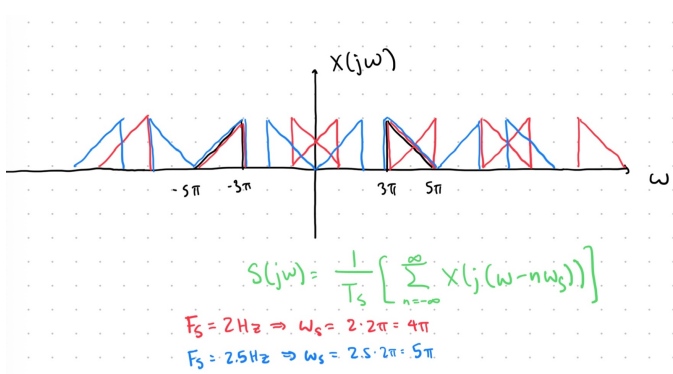
(b) (15 points) Since the signal might have high frequency components, Nyquist rate for this signal can be high. In other words, we need to have a lot of samples of the signal, which means that the sampling scheme is costly. It turns out that for this type of signal, we can sample it at a sampling frequency lower than the Nyquist rate and we can still recover the signal, however in this case, we will use a bandpass filter. To see this, we have the following two options for the sampling frequency:

- $F_s = 2$ Hz;
- $F_s = 2.5$ Hz;

For each case, draw the spectrum of the signal after sampling it. To recover the signal, which F_s can we use? How should we choose the frequencies of the bandpass filter? What is the minimum F_s we can use and still recover the signal?

Solution:

If we use $F_s = 2$ Hz, then $\omega_s = 4\pi$ (shown in red). In this case, we have aliasing. If we sample at $F_s = 2.5$ Hz, then $\omega_s = 5\pi$ (shown in blue). Here, we don't have aliasing. We would want to use a bandpass filter that recovers frequencies between $3\pi \leq \omega \leq 5\pi$. Afterwards, we would use modulation and inverse Fourier Transform to recover the signal.



The minimum F_s that we can use and still recover the signal is 2.5 Hz.

3. (25 points) Sampling with imperfect sampler

Imperfections in a sampler cause characteristic artifacts in the sampled signal. In this problem we will look at the case where the sample timing is non-uniform, as shown below: The sampling function $f(t)$ has its odd samples delayed by a small time τ.

(a) (5 points) Write an expression for $f(t)$ in terms of two uniformly spaced sampling functions.
Solution:

$$f(t) = x_e(t) + x_o(t) \tag{6}$$

where $x_e(t)$ represents the even samples and $x_o(t)$ represents the odd samples. We have

$$x_e(t) = \sum_{n=-\infty}^{\infty} \delta(t-2n) \quad \text{and} \quad x_o(t) = \sum_{n=-\infty}^{\infty} \delta(t-(1+\tau+2n)) \tag{7}$$

Then

$$\begin{aligned} f(t) &= \sum_{n=-\infty}^{\infty} \delta(t-2n) + \sum_{n=-\infty}^{\infty} \delta(t-(1+\tau+2n)) && (8)\\ &= \sum_{n=-\infty}^{\infty} \delta(t-2n) + \delta(t-(1+\tau+2n)) && (9)\\ &= \delta_2(t) + \delta_2(t-(1+\tau)) && (10) \end{aligned}$$

(b) (5 points) Find $F(j\omega)$, the Fourier transform of $f(t)$. Express the impulse trains as sums, and simplify.
Solution:

Recall for an Impulse Train that $c_k = \frac{1}{T}$ for all k and

$$\begin{aligned} f(t) &= \sum_{k=-\infty}^{\infty} c_k e^{j\omega_0 kt} && (11)\\ &= \frac{1}{T} \sum_{n=-\infty}^{\infty} e^{j\omega_0 nt} && (12) \end{aligned}$$

Then

$$\mathcal{F}[f(t)] = \frac{1}{T}\sum_{n=-\infty}^{\infty} \mathcal{F}[e^{j\omega_0 nt}] \tag{13}$$

$$= \frac{1}{T}\sum_{n=-\infty}^{\infty} 2\pi\delta(\omega - n\omega_0) \tag{14}$$

$$= \omega_0 \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_0) \tag{15}$$

$x_e(t)$ is an impulse train with $T = 2$. Then $\omega_0 = \frac{2\pi}{2} = \pi$ and

$$\mathcal{F}[x_e(t)] = \frac{1}{2}\sum_{n=-\infty}^{\infty} 2\pi\delta(\omega - n\pi) \tag{16}$$

$$= \pi \sum_{n=-\infty}^{\infty} \delta(\omega - n\pi) \tag{17}$$

$x_o(t)$ is the same as $x_e(t)$ but with a shift, which introduces an $e^{-j\omega(1+\tau)}$ term. Thus

$$\mathcal{F}[x_o(t)] = e^{-j\omega(1+\tau)}\pi \sum_{n=-\infty}^{\infty} \delta(\omega - n\pi) \tag{18}$$

$$= \pi \sum_{n=-\infty}^{\infty} e^{-j\omega(1+\tau)}\delta(\omega - n\pi) \tag{19}$$

$$= \pi \sum_{n=-\infty}^{\infty} e^{-jn\pi(1+\tau)}\delta(\omega - n\pi) \tag{20}$$

Then

$$F(j\omega) = \pi \sum_{n=-\infty}^{\infty} \delta(\omega - n\pi)\left[1 + e^{-jn\pi(1+\tau)}\right] \tag{21}$$

$$= \pi \sum_{n=-\infty}^{\infty} \delta(\omega - n\pi)\left[1 + (-1)^n e^{-jn\pi\tau}\right] \tag{22}$$

(c) (5 points) Find $F(j\omega)$, for the case where $\tau = 0$, and show that this is what you expect.
Solution:

Plugging in $\tau = 0$, we would just end up with a standard impulse train.

$$F(j\omega) = \pi \sum_{n=-\infty}^{\infty} \delta(\omega - n\pi) \left[1 + e^{-jn\pi}\right] \tag{23}$$

$$= \pi \sum_{n=-\infty}^{\infty} \delta(\omega - n\pi) \left[1 + (e^{j\pi})^{-n}\right] \tag{24}$$

$$= \pi \sum_{n=-\infty}^{\infty} \delta(\omega - n\pi) \left[1 + (-1)^{n}\right] \tag{25}$$

$$\tag{26}$$

If n is odd, then the term becomes zero. Thus n must be even.

$$F(j\omega) = \pi \sum_{n=-\infty}^{\infty} 2\delta(\omega - 2n\pi) \tag{27}$$

$$= 2\pi \sum_{n=-\infty}^{\infty} \delta(\omega - 2n\pi) \tag{28}$$

$$= 2\pi\delta_{2\pi}(\omega) \tag{29}$$

This is a standard impulse train with $T = 1$ and $\omega_0 = 2\pi$, which is what we should expect when $\tau = 0$.

(d) (10 points) Assume the signal we are sampling has a Fourier transform. Sketch the Fourier transform of the sampled signal. Include the baseband replica, and the replicas at $\omega = \pm\pi$. Assume that τ is small, so that $e^{j\omega\tau} \approx 1 + j\omega\tau$.

Solution:

Let $s(t)$ be the sampled signal and $g(t)$ be the signal. Then $s(t) = g(t)f(t)$. Then

$$S(j\omega) = \frac{1}{2\pi}[G(j\omega) * F(j\omega)] \tag{30}$$

$$= \frac{1}{2\pi}\left[G(j\omega) * \pi \sum_{n=-\infty}^{\infty} \delta(\omega - n\pi) \left[1 + (-1)^n e^{-jn\pi\tau}\right]\right] \tag{31}$$

$$= \frac{1}{2\pi}\left[\pi \sum_{n=-\infty}^{\infty} G(j\omega) * \delta(\omega - n\pi) \left[1 + (-1)^n e^{-jn\pi\tau}\right]\right] \tag{32}$$

$$\approx \frac{1}{2}\left[\sum_{n=-\infty}^{\infty} G(j(\omega - n\pi))[1 + (-1)^n(1 - jn\pi\tau)]\right] \tag{33}$$

$$= \frac{1}{2}\left[\sum_{n=-\infty}^{\infty} \Delta\left(\frac{\omega - n\pi}{\pi}\right)[1 + (-1)^n(1 - jn\pi\tau)]\right] \tag{34}$$

When $n = 0$,

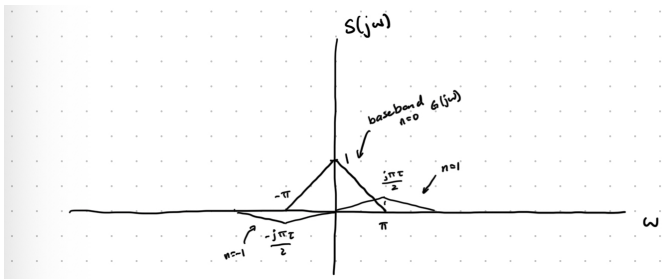
$$S(j\omega) = \frac{1}{2}\Delta\left(\frac{\omega}{\pi}\right)(2) = G(j\omega) \tag{35}$$

When $n = 1$,

$$\frac{1}{2}\Delta\left(\frac{\omega - \pi}{\pi}\right)(1 - 1 + j\pi\tau) = \frac{j\pi\tau}{2}\Delta\left(\frac{\omega - \pi}{\pi}\right) \tag{36}$$

When $n = -1$,

$$\frac{1}{2}\Delta\left(\frac{\omega + \pi}{\pi}\right)(1 - 1 - j\pi\tau) = -\frac{j\pi\tau}{2}\Delta\left(\frac{\omega + \pi}{\pi}\right) \tag{37}$$



4. (15 points) Laplace Transform

Find the Laplace transforms of the following signals and determine their region of convergence.

(a) (7 points) $f(t) = te^{-at}(\cos\omega_0 t)^2 u(t)$

Solution:

$$f(t) = te^{-at}\left(\frac{1+\cos(2\omega_0 t)}{2}\right)u(t) \tag{38}$$
$$= \frac{1}{2}te^{-at}u(t) + \frac{1}{2}te^{-at}\cos(2\omega_0 t)u(t) \tag{39}$$
$$= \frac{1}{2}te^{-at}u(t) + \frac{1}{2}te^{-at}\left(\frac{e^{2j\omega_0 t}+e^{-2j\omega_0 t}}{2}\right)u(t) \tag{40}$$
$$= \frac{1}{2}te^{-at}u(t) + \frac{1}{4}te^{-at}\left(e^{2j\omega_0 t}+e^{-2j\omega_0 t}\right)u(t) \tag{41}$$

Taking the Laplace Transform

$$\mathcal{L}[f(t)] = \frac{1}{2}\int_0^\infty te^{-(a+s)t}dt + \frac{1}{4}\int_0^\infty te^{-(a+s-2j\omega_0)t}dt + \frac{1}{4}\int_0^\infty te^{-(a+s+2j\omega_0)t}dt \tag{42}$$

Using integration by parts:

$$\frac{1}{2}\int_0^\infty te^{-(a+s)t}dt = \frac{1}{2}\left[t\cdot\frac{1}{-(a+s)}e^{-(a+s)t}\right]_0^\infty + \frac{1}{2}\int_0^\infty \frac{1}{a+s}e^{-(a+s)t}dt \tag{43}$$
$$= 0 - 0 + \frac{1}{2}\left[\frac{1}{-(a+s)^2}e^{-(a+s)t}\right]_0^\infty \tag{44}$$
$$= \frac{1}{2(a+s)^2} \tag{45}$$

Thus

$$F(s) = \frac{1}{2(a+s)^2} + \frac{1}{4(a+s-2j\omega_0)^2} + \frac{1}{4(a+s+2j\omega_0)^2} \tag{46}$$
$$= \frac{1}{2(a+s)^2} + \frac{1}{4}\left[\frac{(a+s+2j\omega_0)^2+(a+s-2j\omega_0)^2}{(a+s+2j\omega_0)^2(a+s-2j\omega_0)^2}\right] \tag{47}$$
$$= \frac{1}{2(a+s)^2} + \frac{1}{2}\left[\frac{(a+s)^2+(2j\omega_0)^2}{[(a+s)^2-(2j\omega_0)^2]^2}\right] \tag{48}$$
$$= \frac{1}{2(a+s)^2} + \frac{1}{2}\left[\frac{(a+s)^2-4\omega_0^2}{[(a+s)^2+4\omega_0^2]^2}\right] \tag{49}$$
$$\tag{50}$$

ROC: $\Re(s) > -a$

(b) (8 points)

$$f(t) = \begin{cases} 0, & 0 \le t < 1 \\ 1, & 1 \le t < 2 \\ e^{-2(t-3)}, & 2 \le t < 3 \\ 1, & 3 \le t \end{cases}$$

Solution:

$$\begin{aligned}
F(s) &= \int_1^2 e^{-st}dt + \int_2^3 e^{-2(t-3)}e^{-st}dt + \int_3^\infty e^{-st}dt && (51)\\
&= \left[-\frac{1}{s}e^{-st}\right]_1^2 + \int_2^3 e^{-2t+6-st}dt + \left[-\frac{1}{s}e^{-st}\right]_3^\infty && (52)\\
&= \left[-\frac{1}{s}e^{-st}\right]_1^2 + e^6\int_2^3 e^{-(2+s)t}dt + \left[-\frac{1}{s}e^{-st}\right]_3^\infty && (53)\\
&= -\frac{1}{s}e^{-2s} + \frac{1}{s}e^{-s} + e^6\left[\frac{1}{-(2+s)}e^{-(2+s)t}\right]_2^3 + \frac{1}{s}e^{-3s} && (54)\\
&= -\frac{1}{s}e^{-2s} + \frac{1}{s}e^{-s} + e^6\left[\frac{1}{2+s}(e^{-(2+s)2} - e^{-(2+s)3})\right] + \frac{1}{s}e^{-3s} && (55)\\
&= -\frac{1}{s}e^{-2s} + \frac{1}{s}e^{-s} + e^6\left[\frac{1}{2+s}(e^{-4}e^{-2s} - e^{-6}e^{-3s})\right] + \frac{1}{s}e^{-3s} && (56)\\
&= -\frac{1}{s}e^{-2s} + \frac{1}{s}e^{-s} + \frac{1}{2+s}(e^2e^{-2s} - e^{-3s}) + \frac{1}{s}e^{-3s} && (57)
\end{aligned}$$

ROC: $\Re(s) > 0$ (Condition for the $\int_3^\infty e^{-st}$ to converge)

5. (15 points) Inverse Laplace Transform

Find the inverse Laplace transform $f(t)$ for each of the following $F(s)$: ($f(t)$ is a causal signal)

(a) (7 points) $F(s) = \dfrac{s^2}{(s+2)^2}$x

Solution:

Let $s' = s + 2$. Then

$$F(s) = \frac{(s'-2)^2}{s'^2} \tag{58}$$
$$= 1 - \frac{4}{s'} + \frac{4}{s'^2} \tag{59}$$
$$= 1 - \frac{4}{s+2} + \frac{4}{(s+2)^2} \tag{60}$$

Then

$$f(t) = \delta(t) - 4e^{-2t}u(t) + 4te^{-2t}u(t) \tag{61}$$

(b) (8 points) $F(s) = \dfrac{s-2}{s^2-2s-3}$

Solution:

$$F(s) = \frac{s-2}{(s-3)(s+1)} \tag{62}$$
$$= \frac{A}{s-3} + \frac{B}{s+1} \tag{63}$$

$$(s-3)F(s)|_{s=3} = \frac{1}{4} = A \tag{64}$$
$$(s+1)F(s)|_{s=-1} = \frac{-3}{-4} = \frac{3}{4} = B \tag{65}$$

$$\implies F(s) = \frac{1}{4(s-3)} + \frac{3}{4(s+1)} \tag{66}$$
$$\implies f(t) = \frac{1}{4}e^{3t}u(t) + \frac{3}{4}e^{-t}u(t) \tag{67}$$

Since e^{3t} grows exponentially, we would need $\Re(s) > 3$ so the signal becomes absolutely integrable.

6. (15 points) LTI system

Assume a causal LTI system S_1 is described by the following differential equation:

$$\frac{d^2y(t)}{dt^2} + 3\frac{dy(t)}{dt} + 2y(t) = ax(t), \quad y(0) = 0, \quad y'(0) = 0$$

where a is a constant. Moreover, we know that when the input is e^t, the output of the system S_1 is $\frac{1}{2}e^t$.

(a) (5 points) Find the transfer function $H_1(s)$ of the system. (The answer should not be in terms of a, i.e., you should find the value of a).

Solution:

$$\mathcal{L}[y''(t)] + 3\mathcal{L}[y'(t)] + 2\mathcal{L}[y(t)] = a\mathcal{L}[x(t)] \tag{68}$$

$$s^2Y(s) - sy(0) - y'(0) + 3sY(s) - y(0) + 2Y(s) = aX(s) \tag{69}$$

$$(s^2 + 3s + 2)Y(s) = aX(s) \tag{70}$$

Thus

$$H_1(s) = \frac{Y(s)}{X(s)} = \frac{a}{s^2 + 3s + 2} \tag{71}$$

Since we know that an input $x(t) = e^t$ outputs $y(t) = \frac{1}{2}e^t$, We simply calculate their Laplace transforms and plug into the equation.

$$X(s) = \frac{1}{s-1}, \quad Y(s) = \frac{1}{2(s-1)} \tag{72}$$

Then

$$H_1(s) = \frac{1}{2} = \frac{a}{s^2 + 3s + 2} \tag{73}$$

Plugging in the pole $s = 1$, we get

$$a = \frac{(1)^2 + 3(1) + 2}{2} = 3 \tag{74}$$

$$\implies H_1(s) = \frac{3}{s^2 + 3s + 2} \tag{75}$$

(b) (5 points) Find the output $y(t)$ when the input is $x(t) = u(t)$.

Solution:

$X(s) = \frac{1}{s}$. Then

$$Y(s) = \frac{1}{s} \cdot \frac{3}{s^2 + 3s + 2} \tag{76}$$
$$= \frac{3}{s(s+2)(s+1)} \tag{77}$$
$$= \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+2} \tag{78}$$

$$sY(s)|_{s=0} = \frac{3}{(2)(1)} \implies A = \frac{3}{2} \tag{79}$$

$$(s+1)Y(s)|_{s=-1} = \frac{3}{(-1)(1)} \implies B = -3 \tag{80}$$

$$(s+2)Y(s)|_{s=-2} = \frac{3}{(-2)(-1)} \implies C = \frac{3}{2} \tag{81}$$

Then

$$Y(s) = \frac{3}{2s} - \frac{3}{s+1} + \frac{3}{2(s+2)} \tag{82}$$
$$\implies y(t) = \frac{3}{2}u(t) - 3e^{-t}u(t) + \frac{3}{2}e^{-2t}u(t) \tag{83}$$

(c) (5 points) The system S_1 is linearly cascaded with another causal LTI system S_2. The system S_2 is given by the following input-output pair:

S2 input: $u(t) - u(t-1)$

Output: $r(t) - 2r(t-1) + r(t-2)$

Find the overall impulse response.

Solution:

The overall impulse response is the inverse Laplace transform of $H(s) = H_1(s)H_2(s)$. Therefore we simply need to find $H_2(s)$. We are given that

$$\mathcal{L}[r(t) - 2r(t-1) + r(t-2)] = H_2(s)\mathcal{L}[u(t) - u(t-1)] \tag{84}$$
$$\frac{1}{s^2} - 2\frac{e^{-s}}{s^2} + \frac{e^{-2s}}{s^2} = H_2(s)\left(\frac{1}{s} - \frac{e^{-s}}{s}\right) \tag{85}$$
$$\frac{1}{s}(1 - 2e^{-s} + e^{-2s}) = H_2(s)(1 - e^{-s}) \tag{86}$$
$$\frac{(1-e^{-s})^2}{s} = H_2(s)(1 - e^{-s}) \tag{87}$$

Thus

$$H_2(s) = \frac{1 - e^{-s}}{s} \tag{88}$$

Combining with $H_1(s)$:

$$H(s) = H_1(s) \cdot \frac{1 - e^{-s}}{s} \tag{89}$$

$$= \frac{1}{s} H_1(s) - \frac{1}{s} H_1(s) e^{-s} \tag{90}$$

Notice that $H_1(s)$ is equal to $Y(s)$ from part (b). Since e^{-s} represents a delay by 1 in the time domain,

$$h(t) = y(t) - y(t-1) \tag{91}$$

$$= \left(\frac{3}{2} - 3e^{-t} + \frac{3}{2}e^{-2t}\right) u(t) - \left(\frac{3}{2} - 3e^{-(t-1)} + \frac{3}{2}e^{-2(t-1)}\right) u(t-1) \tag{92}$$